

Chapter 0.5 — Regression Readiness Workout

A First-Principles Regression Study Guide

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Chapter 0.5 — Regression Readiness Workout

A one-evening bridge into Chapter 1, with 180+ exercises

Promise. This workbook teaches only the mathematics, graph reading, and numerical habits that Chapter 1 assumes. It is designed for active work: read a small idea, do several short repetitions, check your understanding, and then use the idea in regression.

Separate explanation pages

If you want the teaching text separately from the drills, read Chapter 0.5 — Regression Readiness Explanations first, then return here for practice. The explanations are also available as a matching PDF.

The running context is the same as Chapter 1: fictional microhydro power (MHP) projects in Khyber Pakhtunkhwa (KP). Costs are measured in **million PKR**, distances in **kilometres**, capacity in **kilowatts**, and terrain difficulty on an index from 1 to 5.

This is a bridge, not a new statistics course. You will practise:

1. graphs, coordinates, straight lines, residuals, and contours;
2. algebra needed to expand and rearrange OLS equations;
3. vectors, distance, dot products, perpendicularity, and projection;
4. matrices, transpose, multiplication, column space, rank, and systems;
5. derivatives, partial derivatives, gradients, Hessians, and minima;
6. floating-point comparison and a simple held-out evaluation; and
7. one cumulative readiness challenge.

You do **not** need probability distributions, hypothesis tests, QR decomposition, SVD, regularisation, gradient descent, or causal diagrams yet.

How to use this tonight

There are more exercises than you need for a single evening. That is deliberate: repetition is available wherever you feel weak.

Use three marks:

- **Core** — do this tonight;
- **Repeat** — do this if the idea is not automatic;
- **Stretch** — save it if time is short.

Suggested elapsed-time route:

Block	Work	Time
0	Setup and diagnostic	15 min
1	Graphs and straight lines	55 min
2	Algebra	65 min
Break	Move, drink water, no screen	10 min
3	Vector geometry	60 min
4	Matrices and rank	70 min
Break	Food and rest	20 min
5	Calculus for minimisation	80 min
6	Numerical computing and held-out data	45 min
7	Cumulative challenge and exit check	50 min

That is about 7 hours including breaks. If you have less time, complete only exercises marked **Core**, then do the final exit check. Accuracy matters more than racing through every item.

The attempt rule

For every exercise:

1. write an answer before looking at the key;
2. if wrong, write one sentence explaining the error;
3. redo it without copying;
4. revisit it after 30–60 minutes.

Use a calculator or Python only when the exercise explicitly says so. The point of the short arithmetic drills is to make the notation feel ordinary.

Minimal setup

```
python -m venv .venv
source .venv/bin/activate
python -m pip install numpy matplotlib
```

Create `chapter_0_5_work.py` and begin it with:

```
import numpy as np
import matplotlib.pyplot as plt
```

Block 0 — Fifteen-Minute Diagnostic

Do these without notes. Do not worry if several are unfamiliar; the results tell you where to spend repetitions.

1. **[Core D1]** On a graph, which axis is horizontal?
2. **[Core D2]** In the point $(4, 9)$, what are the x - and y -coordinates?
3. **[Core D3]** Evaluate $3 + 2(5 - 1)$.
4. **[Core D4]** Expand $2(x + 3)$.
5. **[Core D5]** Expand $(a - b)^2$.
6. **[Core D6]** Solve $3x + 2 = 14$.
7. **[Core D7]** Find the length of the vector $(3, 4)$.
8. **[Core D8]** Calculate $(1, 2) \cdot (3, 4)$.
9. **[Core D9]** What does a zero dot product say about two nonzero vectors?
10. **[Core D10]** If X has shape $(8, 3)$ and β has shape $(3,)$, what is the shape of $X\beta$?
11. **[Core D11]** What does matrix rank count informally?
12. **[Core D12]** Differentiate x^2 .
13. **[Core D13]** If $f'(4) = 0$, must $x = 4$ be a minimum? Explain in one clause.

14. [Core D14] Why can `0.1 + 0.2 == 0.3` be false in Python?

15. [Core D15] Why should a model be evaluated on observations not used to fit it?

Score one point per sound answer.

- 13–15: use the Core exercises and move quickly.
- 9–12: do Core plus Repeat exercises in weak blocks.
- 0–8: do every Core and Repeat exercise; postpone Stretch exercises.

Block 1 — Graphs, Coordinates, Lines, Residuals, and Contours

1.1 Axes, coordinates, scales, and units

A two-dimensional graph uses:

- the horizontal *x*-**axis** for an input such as road distance;
- the vertical *y*-**axis** for an outcome such as project cost;
- an ordered pair (*x*, *y*) to locate a point; and
- a scale and unit on each axis.

If one project has road distance 3 km and cost 22 million PKR, its point is (3, 22). The coordinate order matters.

The **origin** is (0, 0). A point can be observed even when the axes do not begin at zero. Always inspect the tick labels before judging a visual difference: a truncated axis can make a small difference look dramatic.

Exercises: coordinate literacy

1. [Core G1] For (7, 31), state the horizontal and vertical coordinates.
2. [Core G2] Write the point for a project 12 km from a road costing 48 million PKR.
3. [Core G3] What point is the origin?
4. [Core G4] Which point is farther right: (2, 50) or (9, 20)?
5. [Core G5] Which point is higher: (8, 25) or (3, 40)?
6. [Repeat G6] Plot by hand: (0, 8), (1, 10), (2, 12), (3, 14).
7. [Repeat G7] If the *x*-axis changes from kilometres to metres, what happens to the coordinate 2 km?
8. [Repeat G8] A cost axis starts at 95 rather than 0. Explain one visual danger.
9. [Stretch G9] Two charts show the same costs. One spans 0–100; the other spans 80–100. Which makes differences appear larger, and why?

1.2 Scatter plots and visible relationships

A **scatter plot** places one point per observation.

- An upward cloud suggests a positive relationship.
- A downward cloud suggests a negative relationship.
- A shapeless cloud suggests no clear visible relationship.
- A curve suggests the relationship may not be well described by one straight line.

A visible relationship is not automatically causal. Road distance and cost may rise together because remote projects also face difficult terrain.

Exercises: reading a point cloud

10. **[Core G10]** Costs rise as road distance rises. Is the visible relationship positive or negative?
11. **[Core G11]** Capacity rises while cost falls. Is the visible relationship positive or negative?
12. **[Core G12]** Points form a nearly horizontal cloud. What relationship is visible?
13. **[Core G13]** Points form a U-shape. Why is one straight line incomplete?
14. **[Repeat G14]** Give one reason an extreme point deserves inspection rather than automatic deletion.
15. **[Repeat G15]** State why an upward scatter plot does not prove that increasing road distance causes higher cost.
16. **[Stretch G16]** Imagine two terrain groups, each with an upward trend, but the combined data slope downward. Name this phenomenon.

1.3 Straight lines: slope and intercept

A line is

$$\hat{y} = b_0 + b_1x.$$

Here:

- $\hat{y}$ (“y-hat”) is the predicted outcome;
- b_0 is the intercept, the predicted outcome at $x = 0$;
- b_1 is the slope, the predicted change in $\hat{y}$ for a one-unit increase in x .

Slope can also be read from two points:

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}.$$

For $\hat{y} = 8 + 4x$, the intercept is 8 million PKR and the slope is 4 million PKR per km.

Worked example

For the points (2, 14) and (5, 26):

$$b_1 = \frac{26 - 14}{5 - 2} = \frac{12}{3} = 4.$$

Using (2, 14):

$$14 = b_0 + 4(2) \quad \Rightarrow \quad b_0 = 6.$$

The line is $\hat{y} = 6 + 4x$.

Exercises: lines

17. [Core G17] In $\hat{y} = 10 + 3x$, identify the intercept and slope with units if x is km and y is million PKR.
18. [Core G18] Predict at $x = 0$, $x = 2$, and $x = 5$ for $\hat{y} = 10 + 3x$.
19. [Core G19] Find the slope through $(1, 7)$ and $(4, 16)$.
20. [Core G20] Find the line through $(0, 5)$ and $(3, 11)$.
21. [Core G21] A line falls by 8 units when x rises by 4. Find its slope.
22. [Repeat G22] Find the slope through $(2, 20)$ and $(6, 12)$.
23. [Repeat G23] A line has slope 5 and passes through $(2, 17)$. Find its intercept.
24. [Repeat G24] Does $\hat{y} = 4x$ include an intercept? What prediction does it force at $x = 0$?
25. [Repeat G25] Convert the slope 4 million PKR per km into million PKR per metre.
26. [Stretch G26] Two points have the same x but different y . Why is the ordinary slope formula undefined?

1.4 Observations, fitted values, and residuals

An observed value is y_i . A fitted or predicted value is $\hat{y}_i$. The residual is

$$e_i = y_i - \hat{y}_i.$$

On a graph, the residual is the **vertical signed distance** from the fitted line to the observed point.

- $e_i > 0$: the point is above the line; the model underpredicted.
- $e_i < 0$: the point is below the line; the model overpredicted.
- $e_i = 0$: the point is on the line.

For a cost of 25 and prediction of 21, the residual is $25 - 21 = 4$ million PKR.

Exercises: residuals

27. [Core G27] Observed cost is 31 and predicted cost is 28. Find and interpret the residual.
28. [Core G28] Observed cost is 20 and predicted cost is 26. Find and interpret the residual.
29. [Core G29] Calculate predictions and residuals for $x = (0, 1, 2)$, $y = (7, 11, 14)$, and $\hat{y} = 6 + 4x$.
30. [Core G30] Which has the larger absolute error: residual -7 or residual 5 ?
31. [Repeat G31] A point lies 3 units below a fitted line. What is its residual?
32. [Repeat G32] If every residual is positive, what systematic problem does the line have on these observations?
33. [Repeat G33] Why is a horizontal distance not the OLS residual used in Chapter 1?
34. [Stretch G34] Sketch two different lines through the same scatter plot. Mark the residuals for one observation under both lines.

1.5 Contour lines

Imagine the squared-error value depends on two adjustable parameters, b_0 and b_1 . A **contour line** joins parameter pairs with the same error, just as a map contour joins places with the same elevation.

In an OLS contour plot:

- the horizontal and vertical axes are parameters, not observations;
- each loop is one equal-error level;
- moving toward smaller nested loops usually moves toward lower error; and
- the centre of elliptical loops is the minimum.

A long, narrow valley means many parameter pairs give similar error. A perfectly flat direction signals non-uniqueness.

Exercises: contours and graph synthesis

35. **[Core G35]** What quantity stays constant along one error contour?
36. **[Core G36]** In a contour plot with axes b_0 and b_1 , does one dot represent a project or a parameter pair?
37. **[Core G37]** Where is the minimum in a set of nested elliptical contours?
38. **[Repeat G38]** What does a long, narrow contour valley suggest?
39. **[Repeat G39]** Explain the difference between a scatter plot point (x_i, y_i) and an error-surface point (b_0, b_1) .
40. **[Stretch G40]** In one sentence, connect a fitted line in data space to one point in parameter space.

Code lab G

Type, run, and then change the slope from 4 to 2 before rerunning.

```
x = np.array([0.0, 1.0, 2.0, 3.0])
y = np.array([7.0, 11.0, 14.0, 19.0])
b0, b1 = 6.0, 4.0
y_hat = b0 + b1 * x
residuals = y - y_hat

plt.scatter(x, y, label="observed")
plt.plot(x, y_hat, label="fitted")
for xi, yi, yhi in zip(x, y, y_hat):
    plt.vlines(xi, yhi, yi, colors="tab:red")
plt.xlabel("Road distance (km)")
plt.ylabel("Cost (million PKR)")
plt.legend()
plt.show()

print("predictions:", y_hat)
print("residuals:", residuals)
```

41. **[Core G41]** Before running, predict the four fitted values.
42. **[Core G42]** Confirm that each red segment has magnitude $|e_i|$.
43. **[Repeat G43]** With slope 2, which observations are underpredicted?
44. **[Stretch G44]** Add a horizontal line at zero to a separate residual plot.

Block 1 mastery check

Without notes, explain:

1. what slope, intercept, and residual mean;
2. why graph axes and units must be read before the visual pattern;
3. why association in a scatter plot is not an intervention effect; and
4. what a contour line represents.

If any explanation takes more than two sentences, do G17–G40 again tomorrow.

Block 2 — Algebra for Regression Equations

2.1 Signs, fractions, order of operations, and powers

Use the order:

1. brackets;
2. powers;
3. multiplication and division;
4. addition and subtraction.

Remember:

$$-(-a) = a, \quad a^2 = a \cdot a, \quad \sqrt{a^2} = |a|.$$

The square of a negative number is positive: $(-3)^2 = 9$. But $-3^2 = -(3^2) = -9$ unless brackets make the negative part of the base.

Exercises: arithmetic fluency

1. **[Core A1]** Evaluate $5 - 8$.
2. **[Core A2]** Evaluate $-4 - (-7)$.
3. **[Core A3]** Evaluate $3 + 2(5 - 1)$.
4. **[Core A4]** Evaluate $(3 + 2)^2 - 4$.
5. **[Core A5]** Evaluate $(-3)^2$ and -3^2 .
6. **[Core A6]** Simplify $12/18$.
7. **[Repeat A7]** Evaluate $2 - \{3 - [4 - 6]\}$.
8. **[Repeat A8]** Evaluate $|-5| + \sqrt{16}$.
9. **[Repeat A9]** Evaluate $(10 - 4)^2/3$.
10. **[Stretch A10]** If $a = -2$, evaluate $3a^2 - 4a + 1$.

2.2 Distributing and collecting like terms

The distributive law is

$$a(b + c) = ab + ac.$$

It works with subtraction:

$$a(b - c) = ab - ac.$$

Only like terms combine. For example, $3x + 2x = 5x$, but $3x + 2$ cannot be reduced.

Exercises: distribution

11. [Core A11] Expand $3(x + 4)$.
12. [Core A12] Expand $5(2x - 3)$.
13. [Core A13] Expand $-2(x - 6)$.
14. [Core A14] Simplify $3x + 5x - 2$.
15. [Core A15] Simplify $4(x + 2) + 3x$.
16. [Repeat A16] Simplify $2(3x - 1) - 4(x + 5)$.
17. [Repeat A17] Simplify $a(b + c) - ab$.
18. [Repeat A18] Expand $-3(2a - b + 4)$.
19. [Stretch A19] Simplify $2(x - y) - 3(y - x)$.

2.3 Expanding squares

These patterns matter because OLS squares errors:

$$(a - b)^2 = a^2 - 2ab + b^2,$$

$$(a + b)^2 = a^2 + 2ab + b^2.$$

The middle term is essential. $(a - b)^2$ is **not** $a^2 - b^2$.

For one residual:

$$(y - bx)^2 = y^2 - 2ybx + b^2x^2.$$

Exercises: squared expressions

20. [Core A20] Expand $(x - 3)^2$.
21. [Core A21] Expand $(x + 5)^2$.
22. [Core A22] Expand $(y - bx)^2$.
23. [Core A23] Expand $(7 - 2b)^2$.
24. [Core A24] Check A23 by substituting $b = 1$ into both forms.
25. [Repeat A25] Expand $(a - 3c)^2$.
26. [Repeat A26] Expand $(2x + 4)^2$.
27. [Repeat A27] Explain the error in $(a - b)^2 = a^2 - b^2$.
28. [Stretch A28] Expand and simplify $(y - b_0 - b_1x)^2$.

2.4 Rearranging and isolating an unknown

An equation is a balance. Perform the same operation on both sides.

Example:

$$3x + 5 = 20$$

Subtract 5:

$$3x = 15.$$

Divide by 3:

$$x = 5.$$

“Moving a term across” is shorthand, not magic: its sign changes because you add or subtract it on both sides.

Exercises: solve and rearrange

29. **[Core A29]** Solve $x + 7 = 12$.
30. **[Core A30]** Solve $4x = 28$.
31. **[Core A31]** Solve $3x - 5 = 16$.
32. **[Core A32]** Solve $5 - 2x = 11$.
33. **[Core A33]** Solve $ax = c$ for x , assuming $a \neq 0$.
34. **[Core A34]** Rearrange $y = b_0 + b_1x$ to isolate b_0 .
35. **[Repeat A35]** Rearrange $y = b_0 + b_1x$ to isolate x , assuming $b_1 \neq 0$.
36. **[Repeat A36]** Solve $2(x + 3) = 14$.
37. **[Repeat A37]** Solve $3x/4 = 9$.
38. **[Stretch A38]** Solve $(x - 2)/5 = (x + 6)/9$.

2.5 Simultaneous equations

Two unknowns require two independent equations. Use substitution or elimination.

Example:

$$b_0 + b_1 = 5, \quad b_0 + 2b_1 = 8.$$

Subtract the first equation from the second:

$$b_1 = 3.$$

Substitute back: $b_0 + 3 = 5$, so $b_0 = 2$.

Exercises: systems

39. **[Core A39]** Solve $a + b = 9$ and $a - b = 3$.
40. **[Core A40]** Solve $b_0 + b_1 = 7$ and $b_0 + 3b_1 = 15$.
41. **[Core A41]** Solve $2x + y = 11$ and $x + y = 7$.
42. **[Repeat A42]** Solve $3a + 2b = 12$ and $a + 2b = 8$.
43. **[Repeat A43]** Show that $x + y = 4$ and $2x + 2y = 8$ do not determine one unique pair.
44. **[Stretch A44]** Interpret A43 geometrically as two equations describing lines.

2.6 Summations and OLS-style algebra

The sigma symbol means repeated addition:

$$\sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n.$$

Useful rules are:

$$\sum_i (a_i + b_i) = \sum_i a_i + \sum_i b_i,$$

$$\sum_i c a_i = c \sum_i a_i,$$

where c does not change with i .

For the no-intercept loss

$$S(b) = \sum_{i=1}^n (y_i - b x_i)^2,$$

expansion gives

$$S(b) = \sum_i y_i^2 - 2b \sum_i x_i y_i + b^2 \sum_i x_i^2.$$

Exercises: sigma fluency

45. **[Core A45]** Expand $\sum_{i=1}^3 x_i$ without sigma notation.
46. **[Core A46]** For $x = (2, 4, 6)$, calculate $\sum_i x_i$ and $\sum_i x_i^2$.
47. **[Core A47]** For $x = (1, 2)$ and $y = (3, 5)$, calculate $\sum_i x_i y_i$.
48. **[Core A48]** Expand $\sum_i (y_i - b x_i)^2$ using the pattern above.
49. **[Repeat A49]** Why may b move outside $\sum_i b x_i$?
50. **[Repeat A50]** Why may x_i not generally move outside $\sum_i x_i y_i$?
51. **[Repeat A51]** For $x = (1, 2)$, $y = (2, 5)$, write $S(b)$ as a quadratic in b .
52. **[Stretch A52]** Evaluate the A51 quadratic at $b = 2$ and verify it by calculating residuals directly.

Code lab A

```
x = np.array([1.0, 2.0])
y = np.array([2.0, 5.0])

for b in [0.0, 1.0, 2.0, 3.0]:
    direct = np.sum((y - b * x) ** 2)
    expanded = (
        np.sum(y ** 2)
        - 2 * b * np.sum(x * y)
        + b ** 2 * np.sum(x ** 2)
    )
    print(b, direct, expanded)
    assert np.isclose(direct, expanded)
```

53. **[Core A53]** Predict the output for `b=2.0` before running.
54. **[Repeat A54]** Add `b=2.4`. Which tested value has the smallest loss?
55. **[Stretch A55]** Explain why numerical agreement supports the algebra but is not a proof for every possible dataset.

Block 2 mastery check

On blank paper:

1. expand $(y - bx)^2$;
2. solve $4 - 2x = 10$;
3. solve $a + b = 8$ and $a + 3b = 14$; and

4. translate $\sum_{i=1}^3 (y_i - \hat{y}_i)^2$ into ordinary addition.

Do not continue until all four are correct.

Block 3 — Vectors, Length, Angles, and Projection

3.1 A vector is a list and an arrow

A vector such as

$$v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

can mean:

- an ordered list of two numbers;
- an arrow from the origin to $(3, 4)$; or
- a direction and magnitude.

Vector addition and scalar multiplication are componentwise:

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix}$$

$$\begin{bmatrix} a+c \\ b+d \end{bmatrix}, \quad k \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} - k \begin{bmatrix} a \\ b \end{bmatrix}.$$

Exercises: vector operations

1. **[Core V1]** Add $(2, 3) + (4, -1)$.
2. **[Core V2]** Subtract $(5, 2) - (1, 7)$.
3. **[Core V3]** Calculate $3(2, -4)$.
4. **[Core V4]** Find $2u - v$ for $u = (1, 3)$ and $v = (4, 2)$.
5. **[Repeat V5]** Draw $(3, 1)$ and $2(3, 1)$ from the origin. What stays the same?
6. **[Repeat V6]** Draw $v = (2, 4)$ and $-v$. What changes?
7. **[Stretch V7]** Find a and b if $a(1, 0) + b(0, 1) = (7, -2)$.

3.2 Length, distance, and Pythagoras

The Euclidean length or norm of

$$v = (v_1, v_2)$$

is

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

In more dimensions:

$$\|v\| = \sqrt{\sum_i v_i^2}.$$

The distance between a and b is $\|a - b\|$. Squared length is especially important:

$$\|v\|^2 = v^T v = \sum_i v_i^2.$$

OLS minimises the squared residual-vector length.

Exercises: norms and distances

8. **[Core V8]** Find $\|(3, 4)\|$.
9. **[Core V9]** Find $\|(5, 12)\|$.
10. **[Core V10]** Find $\|(1, 2, 2)\|$.
11. **[Core V11]** Find the distance between $(1, 2)$ and $(4, 6)$.
12. **[Core V12]** If residuals are $(3, -4)$, find SSR and residual-vector length.
13. **[Repeat V13]** Compare $\|(6, 8)\|$ with $\|2(3, 4)\|$.
14. **[Repeat V14]** Is $\|(-3, -4)\|$ negative? Explain.
15. **[Repeat V15]** Find all values of a for which $\|(a, 0)\| = 7$.
16. **[Stretch V16]** Verify $\|v\|^2 = v \cdot v$ for $v = (2, -3, 6)$.

3.3 The dot product as arithmetic and geometry

The dot product is

$$u \cdot v = u^T v = \sum_i u_i v_i.$$

It also obeys

$$u \cdot v = \|u\| \|v\| \cos \theta,$$

where θ is the angle between nonzero vectors.

Therefore:

- $u \cdot v > 0$: acute angle, broadly similar direction;
- $u \cdot v < 0$: obtuse angle, broadly opposing direction;
- $u \cdot v = 0$: right angle; the vectors are **orthogonal**.

Exercises: dot products and angles

17. **[Core V17]** Calculate $(1, 2) \cdot (3, 4)$.
18. **[Core V18]** Calculate $(2, -1) \cdot (3, 6)$.
19. **[Core V19]** Show that $(1, 2)$ and $(2, -1)$ are perpendicular.
20. **[Core V20]** Is the angle between $(1, 0)$ and $(1, 1)$ acute, right, or obtuse?
21. **[Core V21]** Is the angle between $(1, 0)$ and $(-2, 1)$ acute, right, or obtuse?
22. **[Repeat V22]** Find a so that $(3, a) \cdot (2, -1) = 0$.
23. **[Repeat V23]** Give a nonzero vector perpendicular to $(4, 7)$.
24. **[Repeat V24]** Explain why a zero dot product cannot by itself define an angle if one vector is the zero vector.
25. **[Stretch V25]** Find $\cos \theta$ for $u = (1, 1)$ and $v = (1, 0)$.

3.4 Projection onto a line

The projection of y onto a nonzero vector x is

$$\text{proj}_x(y)$$

$$\frac{x^T y}{x^T x} x$$

The scalar

$$\hat{b} = \frac{x^T y}{x^T x}$$

tells how much of direction x is needed. The projected vector is $\hat{y} = x\hat{b}$. The residual is $e = y - \hat{y}$, and it is perpendicular to x :

$$x^T e = 0.$$

Worked example

Let $x = (1, 1)$ and $y = (3, 1)$. Then

$$\hat{b} = \frac{(1)(3) + (1)(1)}{1^2 + 1^2} = \frac{4}{2} = 2.$$

So $\hat{y} = (2, 2)$ and $e = (1, -1)$. Check:

$$x^T e = (1)(1) + (1)(-1) = 0.$$

Exercises: projection

26. **[Core V26]** Project $y = (4, 2)$ onto $x = (1, 0)$.
27. **[Core V27]** Project $y = (3, 1)$ onto $x = (1, 1)$.
28. **[Core V28]** For V27, calculate the residual and verify orthogonality.
29. **[Core V29]** Project $y = (2, 5)$ onto $x = (0, 1)$.
30. **[Repeat V30]** Project $y = (5, 1)$ onto $x = (1, 2)$.

31. **[Repeat V31]** For V30, verify that $y = \hat{y} + e$.
32. **[Repeat V32]** Why is projection called the closest point on the line?
33. **[Stretch V33]** If x is multiplied by 10, show that the projected vector does not change.

3.5 The Pythagorean reason projection minimises error

Let $\hat{y}$ be the projection of y onto a line. Any other point q on the line differs from $\hat{y}$ by a vector lying along the line. The residual $y - \hat{y}$ is perpendicular to that difference. Therefore:

$$\|y - q\|^2 = \|y - \hat{y}\|^2 + \|\hat{y} - q\|^2$$

$$\|y - \hat{y}\|^2 + \|\hat{y} - q\|^2 = \|y - q\|^2$$

The second term cannot be negative. Thus no other q is closer than $\hat{y}$.

Exercises: geometric reasoning

34. **[Core V34]** In the Pythagorean identity above, which term is zero only when $q = \hat{y}$?
35. **[Core V35]** Why can adding $\|\hat{y} - q\|^2$ never reduce the distance?
36. **[Core V36]** Translate “the residual is orthogonal to the fitted direction” into a dot-product equation.
37. **[Repeat V37]** Use $y = (3, 1)$, $\hat{y} = (2, 2)$, and $q = (1, 1)$ to verify the identity numerically.
38. **[Repeat V38]** Explain in ordinary language why perpendicularity identifies the nearest point.
39. **[Stretch V39]** Draw the right triangle formed by y , $\hat{y}$, and another attainable q .

Code lab V

```
x = np.array([1.0, 2.0, 3.0])
y = np.array([2.0, 2.0, 5.0])

b_hat = (x @ y) / (x @ x)
y_hat = x * b_hat
e = y - y_hat

print("b_hat:", b_hat)
print("y_hat:", y_hat)
print("residual:", e)
print("orthogonality:", x @ e)
assert np.isclose(x @ e, 0.0)
```

40. **[Core V40]** Compute `b_hat` by hand.
41. **[Core V41]** Why is `np.isclose` used rather than `==`?
42. **[Repeat V42]** Replace `y` with `2*x`. Predict the residual before running.
43. **[Stretch V43]** Test three other values of `b` and confirm none has smaller SSR.

Block 4 — Matrices, Column Space, Rank, and Linear Systems

4.1 Matrix shape and transpose

A matrix is a rectangular table. If X has n rows and p columns, its shape is $n \times p$.

In regression:

- rows are observations;
- columns are features;
- y contains one target per row;
- β contains one coefficient per design column.

The transpose X^T swaps rows and columns. If X is $n \times p$, then X^T is $p \times n$.

Exercises: shapes and transpose

1. **[Core M1]** State the shape of a matrix with 8 observations and 3 features.
2. **[Core M2]** If X is 8×3 , what is the shape of X^T ?
3. **[Core M3]** Transpose $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.
4. **[Core M4]** If β has one coefficient for each of 4 design columns, what is its shape as a one-dimensional NumPy array?
5. **[Repeat M5]** Explain the difference between NumPy shapes $(3,)$, $(3, 1)$, and $(1, 3)$.
6. **[Repeat M6]** If one intercept column is added to 5 features, how many design columns are there?
7. **[Stretch M7]** Why must the order of feature columns at prediction match the order used during fitting?

4.2 Matrix–vector multiplication

For

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ x_{31} & x_{32} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix},$$

the product is

$$X\beta = \begin{bmatrix} x_{11}\beta_1 + x_{12}\beta_2 \\ x_{21}\beta_1 + x_{22}\beta_2 \\ x_{31}\beta_1 + x_{32}\beta_2 \end{bmatrix}.$$

Each output is a row–coefficient dot product. Inner dimensions must match:

$$(n \times p)(p \times 1) = (n \times 1).$$

Exercises: products and predictions

8. [Core M8] Calculate $\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix}$.
9. [Core M9] Calculate predictions for $X = \begin{bmatrix} 1 & 0 & 1 & 2 & 1 & 5 \end{bmatrix}$ and $\beta = (10, 3)^T$.
10. [Core M10] In M9, what role does the first column play?
11. [Core M11] Can a 4×3 matrix multiply a length-4 coefficient vector on the right? Why?
12. [Repeat M12] State the output shape of $(12 \times 4)(4 \times 1)$.
13. [Repeat M13] Expand the second row of $\begin{bmatrix} 1 & 2 & 3 & 1 & 4 & 5 \end{bmatrix} \beta$ term by term.
14. [Repeat M14] Write a plain Python loop that computes each row dot product.
15. [Stretch M15] Explain why elementwise multiplication `X * beta` is not the same object as `X @ beta`.

4.3 Matrix–matrix multiplication

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$. Entry (i, j) is row i of A dotted with column j of B .

Exercises: matrix products

16. [Core M16] State whether $(3 \times 2)(2 \times 4)$ is valid and give the output shape.
17. [Core M17] State whether $(3 \times 2)(3 \times 4)$ is valid.
18. [Core M18] Calculate $\begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 4 & 2 \end{bmatrix}$.
19. [Repeat M19] Show with a small example that generally $AB \neq BA$.
20. [Repeat M20] If X is $n \times p$, state the shapes of $X^T X$ and $X^T y$.
21. [Stretch M21] Why is $X^T X$ always square?

4.4 Linear combinations, span, and column space

Multiplying $X\beta$ forms a linear combination of the columns of X :

$$X\beta = \beta_1 X_{\cdot 1} + \cdots + \beta_p X_{\cdot p}.$$

The **span** of some vectors is every vector obtainable from their linear combinations. The **column space** of X , written $\mathcal{C}(X)$, is the span of its columns.

Every fitted vector $\hat{y} = X\hat{\beta}$ lies in $\mathcal{C}(X)$. OLS selects the point in that space closest to y .

Exercises: span and column space

22. [Core M22] Write $X\beta$ as a combination of the columns when $X = [c_1 \ c_2]$ and $\beta = (3, -2)^T$.
23. [Core M23] Is $(5, 0)$ in the span of $(1, 0)$? If so, give the multiplier.
24. [Core M24] Is $(2, 3)$ in the span of $(1, 0)$?
25. [Core M25] Show that $(5, 7)$ is in the span of $(1, 0)$ and $(0, 1)$.
26. [Repeat M26] Describe the span of one nonzero vector in $\mathbb{R}^2$.
27. [Repeat M27] Describe the span of two nonparallel vectors in $\mathbb{R}^2$.
28. [Repeat M28] Why must $\hat{y} = X\hat{\beta}$ belong to the column space?
29. [Stretch M29] With an intercept and one feature for three observations, which two vectors span the attainable predictions?

4.5 Independence and rank

Columns are linearly independent when no column can be constructed exactly from the others. Matrix **rank** counts independent directions.

For a design matrix with p columns:

- full column rank means $\text{rank}(X) = p$;
- rank deficiency means $\text{rank}(X) < p$;
- full column rank makes $X^T X$ invertible and the OLS coefficient vector unique.

Example of dependence:

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

The second column is twice the first, so the rank is 1, not 2.

Exercises: independence and rank

30. **[Core M30]** Find the rank of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
31. **[Core M31]** Find the rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.
32. **[Core M32]** Is a column of metres independent of the same distance column in kilometres?
33. **[Core M33]** A design has 3 columns and rank 2. Is the coefficient vector unique?
34. **[Core M34]** Can an $n \times p$ matrix have rank greater than $\min(n, p)$?
35. **[Repeat M35]** Give a 3×2 full-column-rank matrix.
36. **[Repeat M36]** Make M35 rank deficient by changing only its second column.
37. **[Repeat M37]** Explain why a numerical solver can return one answer even when coefficients are not uniquely identified.
38. **[Stretch M38]** Explain why duplicate information harms coefficient interpretation even when predictions remain possible.

4.6 Linear systems and inverses

A system can be written

$$A\beta = c.$$

If square A has an inverse,

$$A^{-1}A = I,$$

then

$$\beta = A^{-1}c.$$

An inverse “undoes” a linear transformation. Not every square matrix has one. A matrix with dependent columns is singular and not invertible.

Chapter 1 derives

$$(X^T X)\hat{\beta} = X^T y.$$

These are the normal equations. Although the symbolic solution is

$$\hat{\beta} = (X^T X)^{-1} X^T y,$$

application code should use a least-squares solver, not explicitly compute the inverse.

Exercises: systems and normal equations

39. **[Core M39]** Solve $\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 5 \\ 8 \end{bmatrix}$.
40. **[Core M40]** What does $I\beta$ equal?
41. **[Core M41]** Why does dependent-column matrix X make $X^T X$ non-invertible?
42. **[Core M42]** State the normal equations.
43. **[Repeat M43]** For $X = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $y = (2, 2, 5)^T$, calculate $X^T X$ and $X^T y$.
44. **[Repeat M44]** Solve the one-parameter normal equation from M43.
45. **[Stretch M45]** Distinguish the mathematical closed form from the numerically preferred implementation.

4.7 OLS geometry with several columns

The fitted vector is the projection of y onto $\mathcal{C}(X)$. The residual

$$e = y - X\hat{\beta}$$

is perpendicular to **every design column**:

$$X^T e = 0.$$

When the design includes a column of ones, one row of this equation says

$$\sum_i e_i = 0.$$

Exercises: multi-column geometry

46. **[Core M46]** If X has 3 columns, how many orthogonality conditions are in $X^T e = 0$?
47. **[Core M47]** Explain why $X^T e = 0$ implies each design column has zero dot product with the residual.
48. **[Core M48]** What special residual property follows from an intercept column?
49. **[Repeat M49]** If $e = (1, -2, 1)$, verify it is perpendicular to the intercept vector $(1, 1, 1)$.
50. **[Repeat M50]** Let $X = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 2 & 1 \end{bmatrix}$ and $e = (1, -2, 1)^T$. Verify $X^T e = 0$.
51. **[Stretch M51]** Explain why satisfying $X^T e = 0$ locates the least-squares projection when the objective is convex.

Code lab M

```
X = np.array([
    [1.0, 0.0],
    [1.0, 1.0],
    [1.0, 2.0],
    [1.0, 3.0],
])
y = np.array([7.0, 11.0, 14.0, 19.0])
```

```

beta_hat, _, rank, singular_values = np.linalg.lstsq(X, y, rcond=None)
y_hat = X @ beta_hat
e = y - y_hat

print("shape X:", X.shape)
print("beta:", beta_hat)
print("rank:", rank)
print("X.T @ e:", X.T @ e)
assert rank == X.shape[1]
assert np.allclose(X.T @ e, 0.0, atol=1e-10)

```

52. **[Core M52]** Predict the shapes of `beta_hat`, `y_hat`, and `X.T @ e`.
53. **[Core M53]** Explain both assertions in words.
54. **[Repeat M54]** Add a third column `2 * X[:, 1]`. Predict the new rank.
55. **[Stretch M55]** Run the deficient design and explain why the two slope-like coefficients should not be interpreted separately.

Block 5 — Calculus for Finding the OLS Minimum

5.1 A derivative is local slope

For a function $f(x)$, the derivative

$$f'(x) = \frac{df}{dx}$$

describes the instantaneous rate of change.

- $f'(x) > 0$: the function rises locally as x increases.
- $f'(x) < 0$: the function falls locally.
- $f'(x) = 0$: the graph is locally flat; this is a stationary point.

Useful rules:

$$\begin{aligned} \frac{d}{dx}c &= 0, & \frac{d}{dx}x^k &= kx^{k-1}, \\ \frac{d}{dx}[af(x)] &= af'(x), & \frac{d}{dx}[f(x) + g(x)] &= f'(x) + g'(x). \end{aligned}$$

Exercises: derivative rules

1. **[Core C1]** Differentiate 7.
2. **[Core C2]** Differentiate x .
3. **[Core C3]** Differentiate x^2 .
4. **[Core C4]** Differentiate x^3 .
5. **[Core C5]** Differentiate $5x^2$.
6. **[Core C6]** Differentiate $3x^2 + 4x - 8$.

7. **[Repeat C7]** Differentiate $2x^4 - 3x^2 + 6$.
8. **[Repeat C8]** Evaluate the derivative of x^2 at $x = -3$ and interpret the sign.
9. **[Stretch C9]** Find where $f(x) = x^2 - 6x + 5$ is stationary.

5.2 Chain rule for squared error

If

$$L(b) = [g(b)]^2,$$

the chain rule gives

$$\frac{dL}{db} = 2g(b)g'(b).$$

For one squared residual:

$$L(b) = (y - bx)^2,$$

so

$$\frac{dL}{db} = 2(y - bx)(-x) = -2x(y - bx).$$

The negative sign comes from differentiating $y - bx$ with respect to b .

Exercises: chain rule

10. **[Core C10]** Differentiate $(b - 3)^2$.
11. **[Core C11]** Differentiate $(7 - 2b)^2$.
12. **[Core C12]** Differentiate $(y - bx)^2$ with respect to b .
13. **[Core C13]** In C12, which symbols are constants with respect to b ?
14. **[Repeat C14]** Differentiate $(a + 3b)^2$ with respect to b .
15. **[Repeat C15]** Differentiate $(y - b_0 - b_1x)^2$ with respect to b_0 .
16. **[Repeat C16]** Differentiate the same expression with respect to b_1 .
17. **[Stretch C17]** Expand $(7 - 2b)^2$ first and confirm that ordinary power-rule differentiation matches C11.

5.3 From derivative zero to a minimum

A stationary point is a candidate, not automatically a minimum. The second derivative helps:

- $f''(x) > 0$: curves upward; local minimum;
- $f''(x) < 0$: curves downward; local maximum;
- $f''(x) = 0$: inconclusive.

For $f(x) = (x - 3)^2$:

$$f'(x) = 2(x - 3), \quad f''(x) = 2 > 0.$$

Thus $x = 3$ is the minimum.

Exercises: stationary points

18. **[Core C18]** Find and classify the stationary point of $f(x) = x^2 - 4x + 7$.
19. **[Core C19]** Find and classify the stationary point of $f(x) = -x^2 + 6x$.
20. **[Core C20]** Does $f'(x) = 0$ alone prove a minimum? Give a counterexample.
21. **[Repeat C21]** Find the minimum of $3(b - 5)^2 + 2$.
22. **[Repeat C22]** Complete the square or use derivatives to minimise $b^2 + 8b + 20$.
23. **[Stretch C23]** For $f(x) = x^3$, explain why $f'(0) = 0$ but $x = 0$ is not a minimum.

5.4 Deriving no-intercept OLS

The loss is

$$S(b) = \sum_i (y_i - bx_i)^2.$$

Differentiate:

$$\frac{dS}{db}$$

$$-2 \sum_i x_i (y_i - bx_i).$$

Set it to zero:

$$\sum_i x_i y_i - b \sum_i x_i^2 = 0.$$

Therefore:

$$\hat{b} = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

provided $\sum_i x_i^2 > 0$.

Exercises: scalar OLS

24. **[Core C24]** Use the formula for $x = (1, 2)$ and $y = (2, 5)$.
25. **[Core C25]** Calculate the fitted values and residuals for C24.
26. **[Core C26]** Verify $\sum_i x_i e_i = 0$ for C24.
27. **[Core C27]** What data condition makes $\sum_i x_i^2 = 0$?
28. **[Repeat C28]** Fit no-intercept OLS for $x = (1, 2, 3)$ and $y = (2, 4, 6)$.
29. **[Repeat C29]** Fit it for $x = (1, 2, 3)$ and $y = (1, 1, 1)$.
30. **[Repeat C30]** Differentiate the expanded loss from A51 and solve for its minimum.
31. **[Stretch C31]** Explain why the no-intercept line need not have residuals summing to zero.

5.5 Partial derivatives and gradients

When a function has several inputs, a partial derivative changes one input while holding the others fixed.

For

$$f(a, b) = a^2 + 3ab + 2b^2,$$
$$\frac{\partial f}{\partial a} = 2a + 3b, \quad \frac{\partial f}{\partial b} = 3a + 4b.$$

The gradient collects the partial derivatives:

$$\nabla f = \begin{bmatrix} \partial f / \partial a \\ \partial f / \partial b \end{bmatrix}.$$

It points in the direction of steepest local increase. At an unconstrained smooth minimum, $\nabla f = 0$.

Exercises: partials and gradients

32. **[Core C32]** For $f(a, b) = a^2 + b^2$, find both partial derivatives.
33. **[Core C33]** Find ∇f for $f(a, b) = a^2 + 3ab + 2b^2$.
34. **[Core C34]** Evaluate the C33 gradient at $(a, b) = (1, 2)$.
35. **[Core C35]** Find the stationary point of $f(a, b) = (a - 2)^2 + (b + 1)^2$.
36. **[Repeat C36]** Find the gradient of $f(a, b) = 4a^2 - 2ab + b^2$.
37. **[Repeat C37]** Explain why a zero gradient is the multivariable analogue of a horizontal tangent.
38. **[Stretch C38]** Solve $\nabla f = 0$ for $f(a, b) = a^2 + ab + b^2 - 3a$.

5.6 The OLS gradient and normal equations

The matrix loss is

$$S(\beta) = (y - X\beta)^T (y - X\beta).$$

Its gradient is

$$\nabla_{\beta} S = -2X^T (y - X\beta).$$

At a stationary point:

$$-2X^T (y - X\hat{\beta}) = 0.$$

Rearranging gives:

$$X^T X \hat{\beta} = X^T y.$$

Notice that the calculus result and the geometry result are the same: $X^T e = 0$.

Exercises: connect calculus and geometry

39. **[Core C39]** Substitute $e = y - X\hat{\beta}$ into the zero-gradient equation.
40. **[Core C40]** Rearrange $X^T (y - X\hat{\beta}) = 0$ into the normal equations.
41. **[Core C41]** Translate $X^T (y - X\hat{\beta}) = 0$ geometrically.
42. **[Repeat C42]** What are the shapes of the gradient and β ?
43. **[Repeat C43]** Why is the gradient zero at the OLS solution?
44. **[Stretch C44]** Explain why the same condition appears from two different routes—projection and differentiation.

5.7 Hessians and convexity

The Hessian collects second partial derivatives. For OLS:

$$H = 2X^T X.$$

For any vector z :

$$z^T H z = 2z^T X^T X z = 2 \|Xz\|^2 \geq 0.$$

Therefore the OLS loss is convex: there are no deceptive local minima.

- If X has full column rank, $\|Xz\|^2 > 0$ for every nonzero z ; the loss is strictly convex and the minimiser is unique.
- If X is rank deficient, some nonzero z has $Xz = 0$; the loss has a flat direction and coefficients are not unique.

Exercises: curvature and uniqueness

45. **[Core C45]** State the OLS Hessian.
46. **[Core C46]** Why is $\|Xz\|^2$ never negative?
47. **[Core C47]** What condition makes the OLS minimiser unique?
48. **[Core C48]** What shape does a positive second derivative suggest in one dimension?
49. **[Repeat C49]** Connect a rank-deficient design to a flat contour direction.
50. **[Repeat C50]** Distinguish convex from strictly convex in terms of uniqueness here.
51. **[Stretch C51]** For $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, calculate $2X^T X$.

5.8 Finite-difference checking

A central finite difference estimates a derivative:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

For a gradient, nudge one parameter at a time. Finite differences are a check, not a replacement for understanding. If h is too large, the approximation is crude; if it is extremely tiny, floating-point cancellation can dominate.

Exercises: derivative checks

52. **[Core C52]** Use $h = 0.01$ to approximate the derivative of $f(x) = x^2$ at $x = 3$.
53. **[Core C53]** Compare C52 with the exact derivative.
54. **[Repeat C54]** Predict what happens if $h = 1$.
55. **[Repeat C55]** Why is “make h as tiny as possible” not a perfect rule?
56. **[Stretch C56]** Write a Python function that finite-difference checks one component of the OLS gradient.

Code lab C

```
X = np.array([
    [1.0, 0.0],
    [1.0, 1.0],
    [1.0, 2.0],
])
y = np.array([1.0, 2.0, 2.0])
```



```

beta = np.array([0.5, 0.8])

def ssr(beta_value):
    e = y - X @ beta_value
    return e @ e

analytic = -2.0 * X.T @ (y - X @ beta)
numeric = np.zeros_like(beta)
h = 1e-6

for j in range(beta.size):
    step = np.zeros_like(beta)
    step[j] = h
    numeric[j] = (ssr(beta + step) - ssr(beta - step)) / (2 * h)

print("analytic:", analytic)
print("numeric:", numeric)
assert np.allclose(analytic, numeric, rtol=1e-6, atol=1e-8)

```

57. [Core C57] Explain what `step[j] = h` accomplishes.
58. [Core C58] Explain why the numerator uses two loss evaluations.
59. [Repeat C59] Change `h` to `1e-2` and compare.
60. [Stretch C60] Fit `beta_hat`, calculate the analytic gradient there, and explain the result.

Block 6 — Floating-Point Discipline and Held-Out Evaluation

6.1 Why decimal arithmetic is approximate

Computers store most real-valued calculations in finite binary form. Many ordinary decimals cannot be represented exactly. Thus:

```
0.1 + 0.2 == 0.3
```

may be `False`. This is not random; it is a representation limitation.

Use:

```

np.isclose(a, b)
np.allclose(array_a, array_b)

```

Important arguments:

- `atol`: absolute tolerance, useful near zero;
- `rtol`: relative tolerance, useful when magnitudes vary.

Conceptually, closeness is judged by a rule like:

$$|a - b| \leq \text{atol} + \text{rtol}|b|.$$

Exercises: numerical comparison

1. [Core N1] Why is exact `==` risky for calculated floats?
2. [Core N2] Which function compares two scalar floats approximately?
3. [Core N3] Which function compares arrays approximately?
4. [Core N4] Which tolerance is especially important when the expected value is zero?
5. [Repeat N5] Would exact equality be reasonable for two integer row counts? Why?
6. [Repeat N6] Why should tolerances not be made so wide that every result passes?
7. [Repeat N7] Interpret `np.allclose(X.T @ e, 0, atol=1e-10)`.
8. [Stretch N8] Compare `0.1 + 0.2` and `0.3` with `repr` in Python.

6.2 Stable least-squares solving and rank decisions

The closed form is valuable mathematics:

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

But application code should use:

```
beta_hat, residual_sums, rank, singular_values = np.linalg.lstsq(
    X, y, rcond=None
)
```

Why:

- explicit inversion adds avoidable numerical error and work;
- `lstsq` is designed for the least-squares problem;
- it returns rank information;
- nearly dependent columns require a numerical threshold, not only symbolic reasoning.

Full numerical conditioning belongs in Chapter 2. For now, inspect the reported rank and avoid interpreting coefficients from a rank-deficient design.

Exercises: solver literacy

9. [Core N9] Which NumPy function should Chapter 1 application code use to fit OLS?
10. [Core N10] What rank is required for a design with 4 columns to have full column rank?
11. [Core N11] If `rank == 3` for 4 columns, what should happen before interpreting coefficients?
12. [Core N12] Does `lstsq` returning numbers prove each coefficient is meaningful?
13. [Repeat N13] Give two reasons not to explicitly calculate the inverse in application code.
14. [Repeat N14] What do very small singular values warn about informally?
15. [Stretch N15] Create a nearly duplicate column and compare its singular values with an exactly duplicate column.

6.3 Training and held-out data

If you fit and evaluate on the same observations, the evaluation rewards the model for data it already saw. A **held-out set** provides a small test of performance on unseen observations.

For this bridge:

1. shuffle row indices with a fixed random seed;
2. assign, for example, 75% to training and 25% to held-out data;
3. fit all coefficients using training data only;
4. calculate predictions and metrics separately on both sets; and
5. never move held-out rows back into training because their result is disappointing.

For very small or structured data, one random split is unstable. Chapter 2 develops validation, test separation, and more careful evaluation. Tonight's goal is only to prevent direct training-data reuse.

Exercises: split discipline

16. **[Core N16]** Which rows may influence fitted coefficients?
17. **[Core N17]** Which rows estimate performance on unseen data?
18. **[Core N18]** Why set a random seed?
19. **[Core N19]** Name one danger of a single split on a tiny dataset.
20. **[Core N20]** Why is repeatedly checking the held-out result and changing the model a form of leakage?
21. **[Repeat N21]** With 20 rows and a 75/25 split, how many rows are in each set?
22. **[Repeat N22]** Should scaling values such as means be computed before or after splitting? On which set should they be learned?
23. **[Repeat N23]** A rare remote district appears only in the held-out set. What limitation should be reported?
24. **[Stretch N24]** Explain why chronological data may require a time-respecting split rather than random shuffling.

6.4 Metrics from definitions

For m evaluation observations:

$$\text{MSE} = \frac{1}{m} \sum_i e_i^2,$$

$$\text{RMSE} = \sqrt{\text{MSE}},$$

$$\text{MAE} = \frac{1}{m} \sum_i |e_i|.$$

RMSE and MAE have the same unit as y . RMSE reacts more strongly to large errors because errors are squared.

Exercises: held-out metrics

25. **[Core N25]** For residuals $(2, -1, 3)$, calculate MSE, RMSE, and MAE.
26. **[Core N26]** Which metric is more sensitive to one very large residual: RMSE or MAE?
27. **[Core N27]** If cost is million PKR, what are the units of MSE and RMSE?
28. **[Core N28]** Training RMSE is 1.2 and held-out RMSE is 8.9. What does the gap suggest?
29. **[Repeat N29]** Can a low held-out RMSE establish causation? Why?
30. **[Repeat N30]** Why should training and held-out metrics use the same formula?
31. **[Stretch N31]** Construct two residual sets with the same MAE but different RMSE.

Code lab N: a clean held-out split

```
rng = np.random.default_rng(42)

X_features = np.array([
    [2.0, 40.0],
    [4.0, 45.0],
    [5.0, 50.0],
    [7.0, 55.0],
    [9.0, 62.0],
    [11.0, 70.0],
    [13.0, 76.0],
    [15.0, 85.0],
    [17.0, 92.0],
    [20.0, 100.0],
    [22.0, 110.0],
    [25.0, 120.0],
])
y = np.array([18.0, 21.0, 24.0, 28.0, 31.0, 36.0,
              39.0, 44.0, 49.0, 55.0, 59.0, 67.0])

indices = rng.permutation(len(y))
cut = int(0.75 * len(y))
train_idx = indices[:cut]
held_idx = indices[cut:]

X_train_raw = X_features[train_idx]
X_held_raw = X_features[held_idx]
y_train = y[train_idx]
y_held = y[held_idx]

X_train = np.column_stack([np.ones(len(train_idx)), X_train_raw])
X_held = np.column_stack([np.ones(len(held_idx)), X_held_raw])

beta_hat, _, rank, _ = np.linalg.lstsq(X_train, y_train, rcond=None)
if rank < X_train.shape[1]:
    raise ValueError("Training design is rank deficient")

def metrics(y_true, y_pred):
    e = y_true - y_pred
    return {
        "rmse": float(np.sqrt(np.mean(e ** 2))),
        "mae": float(np.mean(np.abs(e))),
    }

print("training:", metrics(y_train, X_train @ beta_hat))
print("held-out:", metrics(y_held, X_held @ beta_hat))
```

32. [Core N32] How many training and held-out rows are created?
33. [Core N33] Identify the only line that learns coefficients.
34. [Core N34] Confirm that held-out targets do not influence `beta_hat`.
35. [Repeat N35] Run twice and explain why the split repeats.
36. [Repeat N36] Change the seed and explain why the metrics change.
37. [Stretch N37] Add a check that both design matrices have the same number of columns.

Block 7 — Cumulative Regression Readiness

Challenge

Do this section without the answer key. It deliberately combines every block.

Dataset

Eight fictional projects have one feature, road distance x in km, and cost y in million PKR:

Project	x	y
A	1	11
B	2	13
C	3	16
D	4	18
E	6	23
F	7	25
G	8	29
H	10	32

Use A–F as training data and G–H as held-out data. This fixed split is for arithmetic practice, not a general recommendation.

Part 1 — See and describe

1. **[Core X1]** Label suitable axes, units, and scales for a scatter plot.
2. **[Core X2]** Plot the eight points by hand or with Matplotlib.
3. **[Core X3]** Describe the visible direction and whether one line seems plausible.
4. **[Core X4]** Explain why the graph does not prove that moving a site farther from a road would cause its cost to rise.

Part 2 — Build the training design

5. **[Core X5]** Write the 6×2 training design matrix with an intercept.
6. **[Core X6]** State the shapes of X_{train} , y_{train} , and β .
7. **[Core X7]** Explain what each design column means.
8. **[Core X8]** Determine whether the design has full column rank and explain without software.

Part 3 — Fit from centred scalar formulas

For one feature with an intercept:

$$\hat{b}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}.$$

9. **[Core X9]** Calculate training $\bar{x}$ and $\bar{y}$.
10. **[Core X10]** Make columns for $x_i - \bar{x}$, $y_i - \bar{y}$, their product, and $(x_i - \bar{x})^2$.
11. **[Core X11]** Calculate $\hat{b}_1$ and $\hat{b}_0$.
12. **[Core X12]** Interpret both coefficients with units and appropriate caution.

Part 4 — Predictions, residuals, and geometry

13. **[Core X13]** Calculate all six training fitted values and residuals.
14. **[Core X14]** Verify that the training residuals sum to approximately zero.
15. **[Core X15]** Verify that $\sum_i x_i e_i$ is approximately zero.
16. **[Core X16]** Explain what X14 and X15 mean as dot products.
17. **[Core X17]** Calculate training RMSE and MAE.

Part 5 — Held-out evaluation

18. **[Core X18]** Predict projects G and H without refitting.
19. **[Core X19]** Calculate held-out residuals, RMSE, and MAE.
20. **[Core X20]** Compare training and held-out results without claiming that two held-out projects are definitive evidence.

Part 6 — Code proof and deliberate break

21. **[Core X21]** Verify your coefficients using `np.linalg.lstsq`.
22. **[Core X22]** Verify `X_train.T @ residuals` with `np.allclose`.
23. **[Core X23]** Add a third design column equal to `1000 * road_distance`.
24. **[Core X24]** inspect the new rank and explain the failure.
25. **[Repeat X25]** Fit the deficient matrix anyway. Compare predictions with the full-rank model and explain why individual distance coefficients are not interpretable.
26. **[Stretch X26]** Draw or compute an error contour grid for b_0 and b_1 .

Part 7 — One-page memo

27. **[Core X27]** Write a memo of no more than 150 words containing:

- the prediction question;
- units and training/held-out split;
- the fitted equation;
- training and held-out RMSE/MAE;
- one limitation of the tiny synthetic dataset;
- a clear statement that association is not causation; and

- one justified next step.

Final Exit Check

Answer aloud without notes. You are ready to start Chapter 1 when you can answer at least 16 of 18 clearly.

1. What are the axes and one point in a scatter plot?
2. What do slope and intercept mean, including units?
3. What is a residual, and what does its sign mean?
4. What does a contour join?
5. Expand $(a - b)^2$.
6. Solve a two-equation, two-unknown system.
7. What are the arithmetic and geometric meanings of the dot product?
8. What is a vector norm?
9. What makes two vectors orthogonal?
10. What does projection have to do with minimum distance?
11. What does $X\beta$ mean row-wise and column-wise?
12. What is the column space?
13. What does full column rank mean?
14. What are the normal equations?
15. What does a derivative measure?
16. Why is a zero gradient not the whole uniqueness argument?
17. Why use `allclose` and `lstsq`?
18. Why evaluate on held-out observations?

If you miss:

- 1–4: repeat Block 1;
- 5–6: repeat Block 2;
- 7–10: repeat Block 3;
- 11–14: repeat Block 4;
- 15–16: repeat Block 5;
- 17–18: repeat Block 6.

Concise Answer Key

Attempt first. A concise answer is provided so you can check quickly; where an explanation is requested, your wording may differ if the reasoning is sound.

Diagnostic

D1 horizontal. D2 $x = 4, y = 9$. D3 11. D4 $2x + 6$. D5 $a^2 - 2ab + b^2$. D6 $x = 4$. D7 5. D8 11. D9 perpendicular/orthogonal. D10 (8,) . D11 independent directions. D12 $2x$. D13 no; it can be a maximum or neither. D14 finite binary representation. D15 to estimate performance on unseen observations.

Block 1

- G1: 7 horizontal, 31 vertical. G2 (12, 48). G3 (0, 0). G4 (9, 20). G5 (3, 40). G7 2000 m. G8 truncated scale exaggerates differences. G9 80–100.
- G10 positive. G11 negative. G12 little/no visible relationship. G13 direction changes across x . G14 it may be genuine, informative, or a data error requiring investigation. G15 association may reflect other variables. G16 Simpson's paradox.
- G17 intercept 10 million PKR; slope 3 million PKR/km. G18 10, 16, 25. G19 3. G20 $\hat{y} = 5 + 2x$. G21 -2 . G22 -2 . G23 7. G24 no; forces 0. G25 0.004 million PKR/m. G26 division by $x_2 - x_1 = 0$.
- G27 3, underprediction. G28 -6 , overprediction. G29 predictions (6, 10, 14); residuals (1, 1, 0). G30 -7 . G31 -3 . G32 systematic underprediction. G33 OLS defines error in the outcome coordinate.
- G35 error/loss. G36 parameter pair. G37 centre/smallest loop. G38 weak identification or highly correlated parameters. G39 one is an observation; the other selects an entire fitted line. G40 each parameter-space point defines one fitted line. G41 (6, 10, 14, 18). G43 all except possibly the first are underpredicted; with slope 2, predictions are (6, 8, 10, 12) and all observed values exceed them.

Block 2

- A1 -3 . A2 3. A3 11. A4 21. A5 9 and -9 . A6 $2/3$. A7 7. A8 9. A9 12. A10 21.
- A11 $3x + 12$. A12 $10x - 15$. A13 $-2x + 12$. A14 $8x - 2$. A15 $7x + 8$. A16 $2x - 22$. A17 ac . A18 $-6a + 3b - 12$. A19 $5x - 5y$.
- A20 $x^2 - 6x + 9$. A21 $x^2 + 10x + 25$. A22 $y^2 - 2ybx + b^2x^2$. A23 $49 - 28b + 4b^2$. A24 both give 25. A25 $a^2 - 6ac + 9c^2$. A26 $4x^2 + 16x + 16$. A27 missing cross-term; subtraction does not distribute through squaring. A28 $y^2 + b_0^2 + b_1^2x^2 - 2yb_0 - 2yb_1x + 2b_0b_1x$.
- A29 5. A30 7. A31 7. A32 -3 . A33 c/a . A34 $b_0 = y - b_1x$. A35 $x = (y - b_0)/b_1$. A36 4. A37 12. A38 7.
- A39 $a = 6, b = 3$. A40 $b_0 = 3, b_1 = 4$. A41 $x = 4, y = 3$. A42 $a = 2, b = 3$. A43 infinitely many pairs on the same line. A44 identical lines.
- A45 $x_1 + x_2 + x_3$. A46 12 and 56. A47 13. A48 $\sum_i y_i^2 - 2b \sum_i x_i y_i + b^2 \sum_i x_i^2$. A49 b is constant across i . A50 x_i varies with i . A51 $S(b) = 29 - 24b + 5b^2$. A52 at $b = 2$, loss 1; residuals (0, 1). A53 1.0 1.0. A54 $b = 2.4$ has loss 0.2, the smallest of those tested.

Block 3

- V1 (6, 2). V2 (4, -5). V3 (6, -12). V4 ($-2, 4$). V5 direction; length doubles. V6 direction reverses, length stays. V7 $a = 7, b = -2$.
- V8 5. V9 13. V10 3. V11 5. V12 SSR 25, length 5. V13 both 10. V14 no; norms are nonnegative. V15 $a = \pm 7$. V16 both equal 49.
- V17 11. V18 0. V19 dot product $2 - 2 = 0$. V20 acute. V21 obtuse. V22 $a = 6$. V23 e.g. (7, -4). V24 the angle formula divides by zero norm. V25 $1/\sqrt{2}$.

- V26 $\hat{y} = (4, 0)$. V27 $\hat{y} = (2, 2)$. V28 $e = (1, -1)$ and dot product 0. V29 $(0, 5)$. V30 multiplier $7/5$; $\hat{y} = (7/5, 14/5)$. V31 $e = (18/5, -9/5)$ and sums correctly. V32 all alternative points add a nonnegative perpendicular squared distance. V33 scaling cancels between numerator, denominator, and vector.
- V34 $\|\hat{y} - q\|^2$. V35 squared norms are nonnegative. V36 $x^T e = 0$. V37 left side 4; right side $2 + 2 = 4$. V40 $21/14 = 1.5$. V42 zero vector.

Block 4

- M1 8×3 . M2 3×8 . M3 $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. M4 $(4,)$. M5 vector, column matrix, row matrix. M6 6. M7 coefficient meaning depends on position.
- M8 $(17, 39)^T$. M9 $(10, 16, 25)^T$. M10 intercept. M11 no; inner dimensions 3 and 4 disagree. M12 12×1 . M13 $\beta_1 + 4\beta_2 + 5\beta_3$. M15 elementwise result retains/broadcasts matrix dimensions rather than producing row dot products.
- M16 valid, 3×4 . M17 invalid. M18 $\begin{bmatrix} 11 & 5 \\ 4 & 2 \end{bmatrix}$. M20 $p \times p$ and $p \times 1$. M21 the same p appears as rows and columns.
- M22 $3c_1 - 2c_2$. M23 yes, multiplier 5. M24 no. M25 $5(1, 0) + 7(0, 1)$. M26 a line through the origin. M27 all of $\mathbb{R}^2$. M28 it is literally a column linear combination. M29 the ones vector and feature vector.
- M30 2. M31 1. M32 no; one is exactly 1000 times the other. M33 no. M34 no. M35 e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. M36 make column 2 twice column 1. M37 a solver can choose one member of infinitely many minimisers. M38 coefficient allocation across duplicate directions is arbitrary/unstable.
- M39 $a = 5, b = 4$. M40 β . M41 dependence provides a nonzero direction mapped to zero. M42 $X^T X \hat{\beta} = X^T y$. M43 14 and 21. M44 $\hat{b} = 1.5$. M45 derive with the inverse; compute with a least-squares solver.
- M46 3. M47 each entry of the product is a column-residual dot product. M48 residuals sum to zero. M49 $1 - 2 + 1 = 0$. M50 both dot products are zero. M52 $(2,)$, $(4,)$, $(2,)$. M53 full rank and residual orthogonality. M54 rank remains 2, not 3.

Block 5

- C1 0. C2 1. C3 $2x$. C4 $3x^2$. C5 $10x$. C6 $6x + 4$. C7 $8x^3 - 6x$. C8 -6 , locally decreasing as x increases. C9 $x = 3$.
- C10 $2(b - 3)$. C11 $-4(7 - 2b) = 8b - 28$. C12 $-2x(y - bx)$. C13 x and y . C14 $6(a + 3b)$. C15 $-2(y - b_0 - b_1x)$. C16 $-2x(y - b_0 - b_1x)$.
- C18 $x = 2$, minimum; $f'' = 2$. C19 $x = 3$, maximum; $f'' = -2$. C20 no; $-x^2$ at zero. C21 $b = 5$, minimum value 2. C22 $b = -4$, minimum 4. C23 the function continues increasing through zero.
- C24 $12/5 = 2.4$. C25 $(2.4, 4.8)$ and $(-0.4, 0.2)$. C26 $-0.4 + 0.4 = 0$. C27 every $x_i = 0$. C28 2 with zero residuals. C29 $6/14 = 3/7$. C30 derivative $10b - 24 = 0$, so $b = 2.4$. C31 without a ones column, orthogonality is to x , not necessarily to ones.
- C32 $2a, 2b$. C33 $(2a + 3b, 3a + 4b)^T$. C34 $(8, 11)^T$. C35 $(2, -1)$. C36 $(8a - 2b, -2a + 2b)^T$. C38 $(2, -1)$.
- C39 $X^T e = 0$. C40 expand to $X^T y - X^T X \hat{\beta} = 0$ and rearrange. C41 residual perpendicular to every design column. C42 same shape, one entry per coefficient. C43 no local first-order change reduces the loss. C44 both describe the same closest point.

- C45 $2X^T X$. C46 it is a sum of squares. C47 full column rank. C48 upward bowl. C49 a nonzero parameter direction does not change predictions/loss. C50 strict convexity guarantees one minimiser; ordinary convexity may have several. C51 $2I$.
- C52 $[(3.01)^2 - (2.99)^2]/0.02 = 6$. C53 exact derivative is 6. C54 still 6 for this quadratic with central difference. C55 cancellation/rounding eventually dominates. C57 nudges only component j . C58 it estimates symmetric change around the current value. C60 gradient is numerically zero.

Block 6

- N1 finite binary approximation. N2 `np.isclose`. N3 `np.allclose`. N4 `atol`. N5 yes; exact counts are integers. N6 loose tests can hide real mistakes. N7 every design-column/residual dot product is within the chosen tolerance of zero.
- N9 `np.linalg.lstsq`. N10 4. N11 reject/flag rank deficiency before coefficient interpretation. N12 no. N13 greater numerical error and unnecessary work; solver also provides diagnostics. N14 near dependence/poor conditioning.
- N16 training rows. N17 held-out rows. N18 reproducibility. N19 result varies greatly with which few rows are selected. N20 the held-out set begins guiding choices and is no longer unseen. N21 15 and 5. N22 after splitting; learn transformations from training only. N23 the result may be dominated by a group the model never saw. N24 future rows must not inform a model evaluated on the past.
- N25 MSE $14/3 \approx 4.667$, RMSE ≈ 2.160 , MAE 2. N26 RMSE. N27 squared million PKR for MSE; million PKR for RMSE. N28 possible overfitting, shift, outliers, or unstable tiny split; investigate. N29 no; predictive performance does not identify interventions. N31 e.g. (0, 2) and (1, 1) both have MAE 1, with RMSE $\sqrt{2}$ and 1.
- N32 9 and 3. N33 the `np.linalg.lstsq` line. N34 only `X_train` and `y_train` enter it. N35 the fixed seed reproduces the permutation. N36 different observations are assigned to each set.

Cumulative challenge

Use software to confirm arithmetic after attempting it by hand.

- X5: $X = \begin{bmatrix} 1 & 1 & 1 & 2 & 1 & 3 & 1 & 4 & 1 & 6 & 1 & 7 \end{bmatrix}$. X6: (6, 2), (6,), (2,). X8 full rank because the distance column is not a constant multiple of ones.
- X9: $\bar{x} = 23/6 \approx 3.8333$ and $\bar{y} = 106/6 \approx 17.6667$.
- X10 numerator $\sum (x_i - \bar{x})(y_i - \bar{y}) = 191/3 \approx 63.6667$; denominator $\sum (x_i - \bar{x})^2 = 161/6 \approx 26.8333$.
- X11 $\hat{b}_1 = 382/161 \approx 2.3727$ and $\hat{b}_0 = 60/7 \approx 8.5714$.
- X12 predicted cost at zero distance is about 8.57 million PKR; each additional km is associated with about 2.37 million PKR higher predicted cost in these training data. This is not a causal effect.
- X13 fitted values are approximately (10.944, 13.317, 15.689, 18.062, 22.807, 25.180); residuals (0.056, -0.317, 0.311, -0.062, 0.193, -0.180).
- X14 and X15 are zero apart from rounding. They express orthogonality to the ones and distance columns.
- X17 training RMSE ≈ 0.213 and MAE ≈ 0.186 million PKR.
- X18 predictions for G and H are approximately 27.553 and 32.298.
- X19 held-out residuals are approximately 1.447 and -0.298; RMSE ≈ 1.045 and MAE ≈ 0.873 million PKR.
- X20 the held-out errors are larger, but two observations cannot establish stable generalisation performance.
- X24 rank is 2 for 3 columns because metres are 1000 times kilometres.

What to Carry Into Chapter 1

Keep these six sentences beside you:

1. A graph is meaningless until I read both axes, scales, and units.
2. A residual is observed minus predicted, and OLS minimises their squared length.
3. $X\beta$ is both one row-wise prediction per observation and a linear combination of design columns.
4. OLS projects y onto the column space, so $X^T e = 0$.
5. Differentiating the squared-error objective produces the same normal equations as the geometry.
6. A computed coefficient is not automatically unique, stable, causal, or useful on unseen data.

You are now prepared to read Chapter 1 in two passes: first for meaning and computation, then for geometry and derivation.